Module 7b slides: PK-PD (worked examples)

Module 7b - PK-PD Relationship · Lecture slides, worked examples · 12 Oct 2026

Name _____ Date _____

The minimum effective concentration (MEC or Ceff) in plasma for a certain antibiotic is 0.2 mcg/mL. The drug follows a one-compartment open model and has an apparent volume of distribution of 24 L and a first-order elimination rate constant of 0.15 h–1.

Slide 7 (Practice Problem)

What is the teff for a single 250-mg IV dose of this antibiotic?

Slide 7 (Practice Problem)

What is the new teff or for this drug if the dose were increased by a factor of four to 1000 mg?

Slide 7 (Practice Problem)

What is the new teff for this drug if the dose were increased by a factor of two to 500 mg?

Slide 8 What dose is required to double teff?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Slide 7 (Practice Problem) Her key

Answer: 26.35 hr

WHAT THE STEM GIVES

Ceff	0.2 mcg/mL	equals 0.2 mg/L, the level where the effect ends
V_D	24 L	divides the dose to give C_0
k	0.15 h–1	the denominator of teff
D_0	250-mg	the numerator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$\text{Ceff} = 0.2 \text{ mcg/mL} = 0.2 \text{ mg/L}$$

1 mcg/mL equals 1 mg/L, because multiplying top and bottom by 1000 turns mcg into mg and mL into L. The dose is in mg and the volume in L, so C_0 comes out in mg/L.

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{250 \text{ mg}}{24 \text{ L}} = 10.42 \text{ mg/L}$$

C_0, the concentration just after an IV bolus, is the whole dose spread through the apparent volume of distribution V_D. The effect lasts while the level falls from here to Ceff.

ALGEBRA

$$\text{teff} = \frac{\ln C_0 - \ln \text{Ceff}}{k} = \frac{\ln(10.42 \div 0.2)}{0.15 \text{ hr}^{-1}} = \frac{\ln 52.08}{0.15} = \frac{3.953}{0.15 \text{ hr}^{-1}}$$

This is $C = C_0 e^{-kt}$ solved for the time at which C reaches Ceff. The stem gives k directly, so the natural-log form needs no half-life; $\ln C_0 - \ln \text{Ceff}$ is the log of their ratio.

ROUND

$$t½ = \frac{0.693}{0.15} = 4.62 \text{ hr}; \text{ teff} = 26.35 \text{ hr}$$

$3.953 \div 0.15 \text{ hr}^{-1} = 26.35$ hr, her value. The ratio 52.08 lies between 2 to the 5th and 2 to the 6th, so the duration is between 5 and 6 half-lives of 4.62 hr.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0 \div \text{Ceff} = 52.08$ lies between 2 to the 5th and 2 to the 6th, so teff is between 5 and 6 half-lives of 4.62 hr: 26.35 hr.

Slide 7 (Practice Problem) Her key

Answer: 35.6 hr

WHAT THE STEM GIVES

D_0	1000 mg	four times 250 mg, so C_0 is four times larger

Ceff	0.2 mcg/mL	equals 0.2 mg/L
V_D	24 L	unchanged by the dose
k	0.15 h–1	unchanged by the dose

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{1000\ \text{mg}}{24\ \text{L}} = 41.67\ \text{mg/L};\ \text{Ceff} = 0.2\ \text{mcg/mL} = 0.2\ \text{mg/L}$$

Four times the dose gives four times C_0, because V_D does not change with the dose. 1 mcg/mL equals 1 mg/L, so Ceff goes in as 0.2 mg/L.

ALGEBRA

$$\text{teff} = \frac{\ln(41.67 \div 0.2)}{0.15\ \text{hr}^{-1}} = \frac{\ln 208.33}{0.15} = \frac{5.3391}{0.15\ \text{hr}^{-1}} = 35.594\ \text{hr}$$

The same natural-log duration line with the new C_0. The ratio $C_0 \div$ Ceff is four times larger, so its logarithm is larger by ln 4 = 1.386, not four times larger.

ROUND

$$35.594 - 26.352 = 9.242\ \text{hr} = 2 \times 4.621\ \text{hr};\ \text{teff} = 35.6\ \text{hr}$$

A factor of four is two doublings, and each doubling of the dose adds one half-life, 4.621 hr, to the 26.352 hr of the 250-mg dose. 35.594 hr is 35.6 hr to one decimal place. Four times the dose gives far less than four times the duration.

> **DOES THE NUMBER MAKE SENSE?**
> Four times the dose adds two half-lives, 2 × 4.621 hr, to the 26.352 hr of part a: 35.6 hr, far short of four times the duration.

Slide 7 (Practice Problem) Her key

Answer: 30.97 hr

WHAT THE STEM GIVES

D_0	500 mg	twice 250 mg, so C_0 doubles
Ceff	0.2 mcg/mL	equals 0.2 mg/L
V_D	24 L	unchanged by the dose
k	0.15 h–1	unchanged by the dose

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{500\ \text{mg}}{24\ \text{L}} = 20.83\ \text{mg/L};\ \text{Ceff} = 0.2\ \text{mcg/mL} = 0.2\ \text{mg/L}$$

Twice the dose gives twice C_0, because V_D, the apparent volume of distribution, does not change with the dose. 1 mcg/mL equals 1 mg/L, so Ceff is 0.2 mg/L.

ALGEBRA

$$\text{teff} = \frac{\ln(20.83 \div 0.2)}{0.15\ \text{hr}^{-1}} = \frac{\ln 104.2}{0.15} = \frac{4.646}{0.15\ \text{hr}^{-1}} = 30.97\ \text{hr}$$

The natural-log duration line with the doubled C_0. Its logarithm grows by ln 2 = 0.693, and 0.693 ÷ 0.15 hr^{-1} is the half-life, 4.62 hr.

ROUND

30.97 − 26.35 = 4.62 hr, one half-life; teff = 30.97 hr

Each time the dose is doubled the duration gains one more half-life, 4.62 hr: 26.35 hr becomes 30.97 hr, not 52.7 hr, because the dose sits inside the logarithm.

DOES THE NUMBER MAKE SENSE?

Doubling the dose adds one half-life, 4.62 hr, to 26.35 hr: 30.97 hr, well short of twice 26.35.

The printed problem asks for a factor of four, 1000 mg; her worked answer changes it to a factor of two, 500 mg, and works both doses.

Slide 8 Her key

Answer: 13 g

WHAT THE STEM GIVES

teff	26.35 hr	doubled to 52.7 hr, the target duration
Ceff	0.2 mcg/mL	equals 0.2 mg/L, the level at the target time
V_D	24 L	multiplies Ceff to give 4.8 mg
k	0.15 h–1	in the exponent k teff

THE WORKING, ONE LINE AT A TIME

SETUP

Target teff = 2 × 26.35 hr = 52.7 hr

Doubling the duration means the level must still be at Ceff 52.7 hours after the dose, so the time goes into the dose line as 52.7 hr.

ALGEBRA

$$\text{Ceff} = \frac{D_0}{V_D}\ e^{-k\ \text{teff}},\ \text{so}\ D_0 = \text{Ceff} \times V_D \times e^{k\ \text{teff}}$$

$C = C_0 e^{-kt}$ with $C_0 = D_0$ over V_D, written at the moment the level reaches Ceff, then multiplied through by $V_D\ e^{k\ \text{teff}}$ to leave the dose on its own.

ALGEBRA

$D_0 = (0.2\ \text{mg/L})(24\ \text{L})\ e^{(0.15)(52.7)} = 4.8\ \text{mg} \times e^{7.905} = 4.8\ \text{mg} \times 2710.8 = 13{,}012\ \text{mg}$

0.2 mcg/mL is 0.2 mg/L; mg/L × L leaves mg. The exponent kt = 0.15 hr^{-1} × 52.7 hr = 7.905 has no units, and $e^{7.905}$ is the factor the level must fall by over 52.7 hours.

ROUND

D_0 = 13,012 mg, rounded to 13 g

Her answer is 13 g, about 52 times the 250-mg dose: the extra 26.35 hours is 5.7 more half-lives of 4.62 hr, and each half-life of duration needs the dose doubled once.

> **DOES THE NUMBER MAKE SENSE?**
> Doubling the dose adds only one half-life, so doubling a 26.35-hr duration needs far more than twice 250 mg: 13 g is more than 50 times it.

Worked with teff = 52.7 hr, as she rounds it; carrying teff unrounded, 52.705 hr, gives 13.02 g. Both are 13 g.
